

Brac University
Department of Computer Science and Engineering

CSE111: Programming Language II Summer 2024 Quiz 3 Marks: 10

Name:	ID:	Section:
-------	-----	----------

1. **Create/design a class called Library in which** the results displayed in the output box are generated. [You are not allowed to change the code below]

[10]

Driver Codes/Tester Class:

```
public class LibraryTester {
    public static void main(String[] args) {
        System.out.println("Total shelf space: " +
Library.getShelfSpace());
        System.out.println("#####");
        Library book1 = new Library("The Great Gatsby", 1.2);
        book1.details();
        book1.addToShelf();
        System.out.println("#####");
        Library book2 = new Library("To Kill a Mockingbird", 1.5);
        book2.addToShelf();
        System.out.println("Available shelf space: " +
Library.getShelfSpace());
        System.out.println("#####");
        Library book3 = new Library("War and Peace", 2.8);
        book3.addToShelf();
        book3.details();
        System.out.println("#####");
        book2.removeFromShelf();
        System.out.println("Available shelf space: " +
Library.getShelfSpace());
        System.out.println("#####");
        book3.addToShelf();
        book2.details();
        System.out.println("Available shelf space: " +
Library.getShelfSpace());
    }
}
```

Outputs:

```
Total shelf space: 5.0
#####
Book ID: 1, Title: The Great Gatsby, Width:
1.2, On Shelf: false
Book 1 added to shelf.
#####
Book 2 added to shelf.
Available shelf space: 2.3
#####
Cannot add book, exceeds shelf space.
Book ID: 3, Title: War and Peace, Width:
2.8, On Shelf: false
#####
Book 2 removed from shelf.
Available shelf space: 3.8
#####
Book 3 added to shelf.
Book ID: 2, Title: To Kill a Mockingbird,
Width: 1.5, On Shelf: false
Available shelf space: 1.0
```